

Introduction to Calculus

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By

Dr. Akshara Makrariya
Assistant Professor (sr.)
SASL – Mathematics
VIT , Bhopal University.

Module : 1 Integration -1

Content

- ❖ Introduction,
- ❖ Integration of Quadratic function, Exponential function,
- ❖ Logarithmic function,
- ❖ Trigonometric Function.
- ❖ Algebra of Integration-
- ❖ Definite and indefinite integrals with properties.
- ❖ Integral of Implicit Function, Explicit Function, Transcendental Function. Sum, Product and division rules of multiple functions.

Integration by Parts

Suppose that u and v are differentiable functions. Then,

$$\int u \, dv = uv - \int v \, du.$$

Integration by Parts for Definite Integrals

Let u and v be differentiable. Then,

$$\int_a^b u(x)v'(x) dx = u(x)v(x) \Big|_a^b - \int_a^b v(x)u'(x) dx.$$

Integration: The Exponential Form

Integrate Exponential Functions

$$\int e^x dx = e^x + C$$

$$\int a e^x dx = e^{ax} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int a^x \ln a dx = a^x + C$$

Integration of Logarithmic function

Integral of $\ln x$

$$\int \ln x \, dx = x \ln x - x + C$$

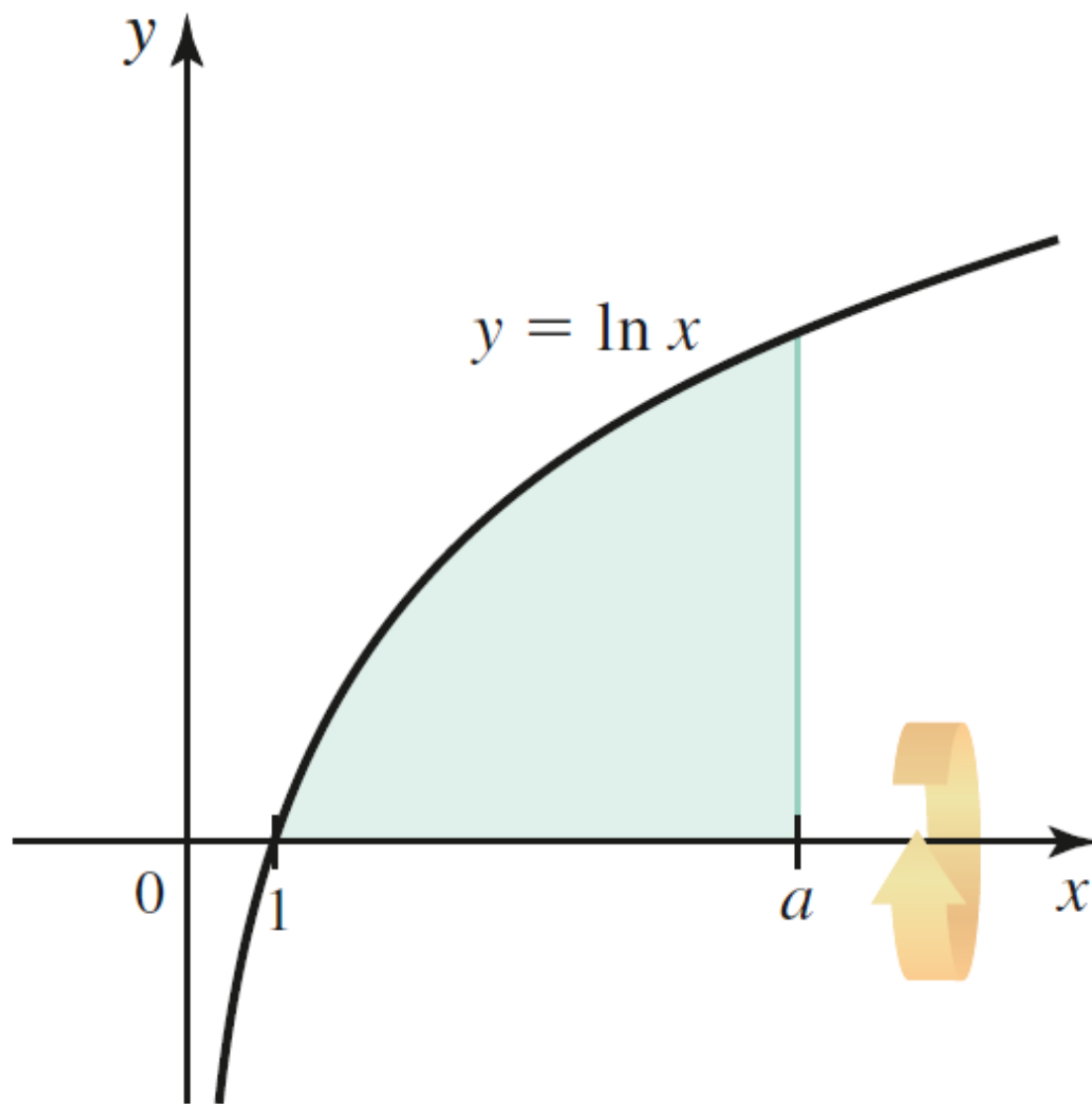


FIGURE 8.1

Trigonometric Integrals

Table 8.1

$$\int \sin^m x \cos^n x \, dx$$

m odd, n real

Strategy

Split off $\sin x$, rewrite the resulting even power of $\sin x$ in terms of $\cos x$, and then use $u = \cos x$.

n odd, m real

Split off $\cos x$, rewrite the resulting even power of $\cos x$ in terms of $\sin x$, and then use $u = \sin x$.

m and n both even,
nonnegative integers

Use half-angle identities to transform the integrand into a polynomial in $\cos 2x$, and apply the preceding strategies once again to powers greater than 1 ($\cos^2 2x$, $\cos^3 2x$, etc.).

THEOREM 8.1 Integrals of $\tan x$, $\cot x$, $\sec x$, and $\csc x$

$$\int \tan x \, dx = -\ln |\cos x| + C = \ln |\sec x| + C \quad \int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C \quad \int \csc x \, dx = -\ln |\csc x + \cot x| + C$$

Reduction Formulas

Assume n is a positive integer.

$$1. \int \sin^n x \, dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

$$2. \int \cos^n x \, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

$$3. \int \tan^n x \, dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \, dx, \quad n \neq 1$$

$$4. \int \sec^n x \, dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx, \quad n \neq 1$$

Table 8.2

$$\int \tan^m x \sec^n x \, dx$$

n even

Strategy

Split off $\sec^2 x$, rewrite the remaining even power of $\sec x$ in terms of $\tan x$, and use $u = \tan x$.

m odd

Split off $\sec x \tan x$, rewrite the remaining even power of $\tan x$ in terms of $\sec x$, and use $u = \sec x$.

m even and n odd

Rewrite the even power of $\tan x$ in terms of $\sec x$ to produce a polynomial in $\sec x$; apply reduction formula 4 to each term.

Trigonometric Substitutions

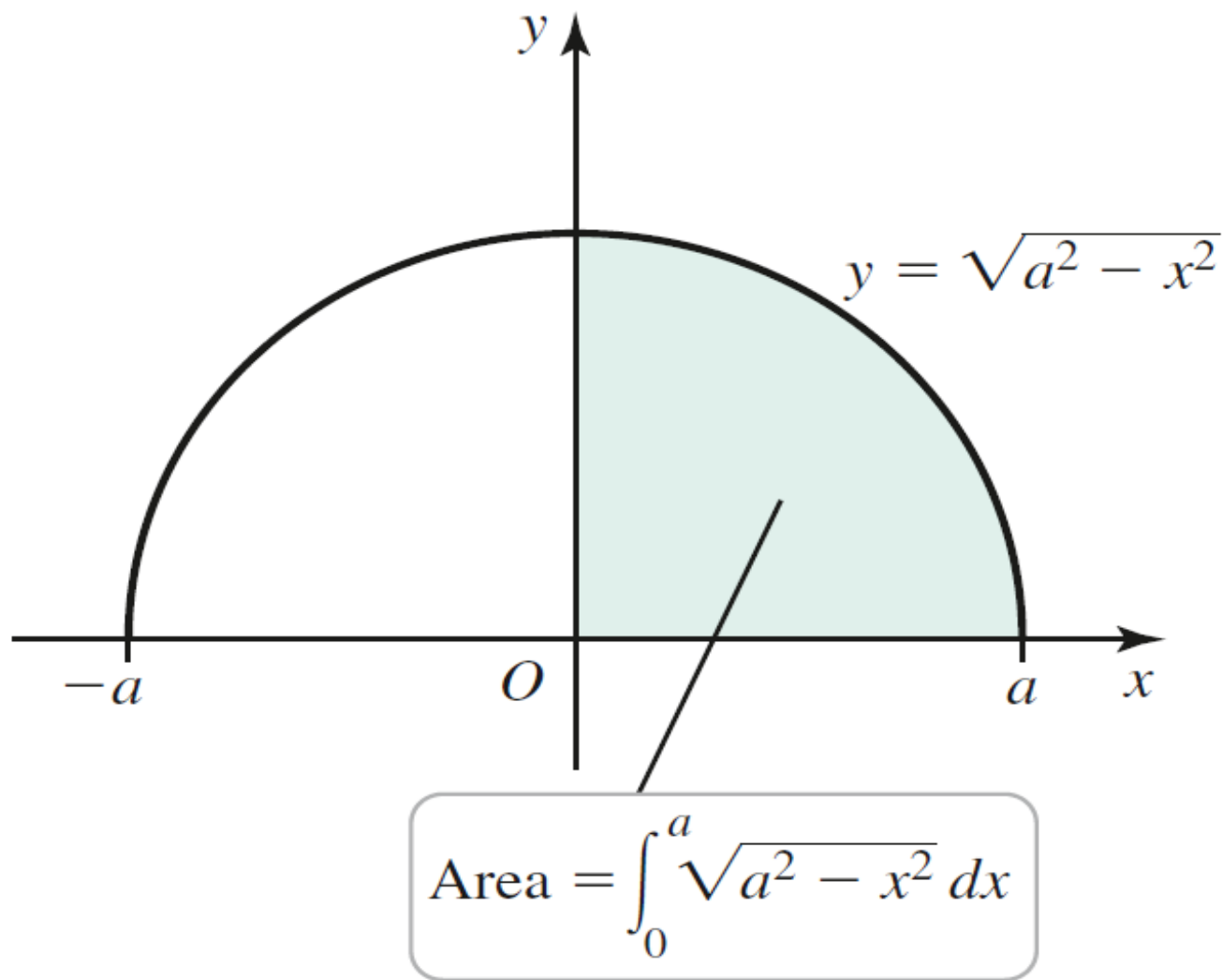


FIGURE 8.2

Trigonometric Substitution

- ❑ In finding the area of a circle or an ellipse, an integral of the form $\int \sqrt{a^2 - x^2} dx$ arises, where $a > 0$.
- ❑ If it were $\int x \sqrt{a^2 - x^2} dx$, the substitution $u = a^2 - x^2$ would be effective but, as it stands, $\int \sqrt{a^2 - x^2} dx$ is more difficult.

Trigonometric Substitution

- If we change the variable from x to θ by the substitution $x = a \sin \theta$, then the identity $1 - \sin^2 \theta = \cos^2 \theta$ allows us to get rid of the root sign because

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2 \theta}$$

$$= \sqrt{a^2(1 - \sin^2 \theta)}$$

$$= \sqrt{a^2 \cos^2 \theta}$$

$$= a |\cos \theta|$$

Example 1

□ Evaluate $\int \frac{\sqrt{9 - x^2}}{x^2} dx$.

□ **Solution:**

Let $x = 3 \sin \theta$, where $-\pi/2 \leq \theta \leq \pi/2$. Then $dx = 3 \cos \theta d\theta$
and

$$\begin{aligned}\sqrt{9 - x^2} &= \sqrt{9 - 9 \sin^2 \theta} \\ &= \sqrt{9 \cos^2 \theta} \\ &= 3 |\cos \theta| \\ &= 3 \cos \theta\end{aligned}$$

□ **(Note that $\cos \theta \geq 0$ because $-\pi/2 \leq \theta \leq \pi/2$.)**

Example 1 – *Solution*

By Inverse Substitution we get:

$$\int \frac{\sqrt{9 - x^2}}{x^2} dx = \int \frac{3 \cos \theta}{9 \sin^2 \theta} 3 \cos \theta d\theta$$

$$= \int \frac{\cos^2 \theta}{\sin^2 \theta} d\theta$$

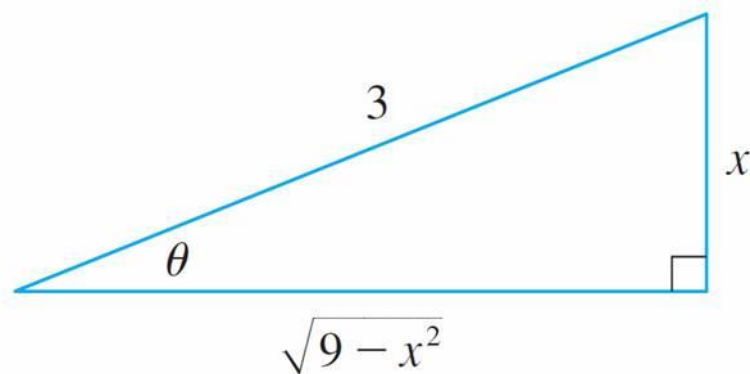
$$= \int \cot^2 \theta d\theta$$

$$= \int (\csc^2 \theta - 1) d\theta$$

$$= -\cot \theta - \theta + C$$

Example 1 – *Solution*

- Since this is an indefinite integral, we must return to the original variable x . This can be done either by using trigonometric identities to express $\cot \theta$ in terms of $\sin \theta = x/3$ or by drawing a diagram, as in Figure 1, where θ is interpreted as an angle of a right triangle.



$$\sin \theta = \frac{x}{3}$$

Example 1 – *Solution*

- Since $\sin \theta = x/3$, we label the opposite side and the hypotenuse as having lengths x and 3 .
- Then the Pythagorean Theorem gives the length of the adjacent side as $\sqrt{9 - x^2}$, so we can simply read the value of $\cot \theta$ from the figure:

$$\cot \theta = \frac{\sqrt{9 - x^2}}{x}$$

Example 1 – *Solution*

- Since $\sin \theta = x/3$, we have $\theta = \sin^{-1}(x/3)$ and so

$$\int \frac{\sqrt{9 - x^2}}{x^2} dx = -\frac{\sqrt{9 - x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + C$$

Table 8.3

**The Integral
Contains...**

Corresponding Substitution

Useful Identity

$$a^2 - x^2$$

$$x = a \sin \theta, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$a^2 - a^2 \sin^2 \theta = a^2 \cos^2 \theta$$

$$a^2 + x^2$$

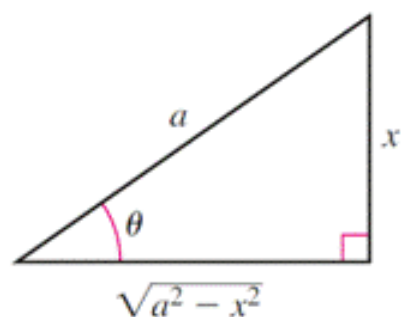
$$x = a \tan \theta, -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$a^2 + a^2 \tan^2 \theta = a^2 \sec^2 \theta$$

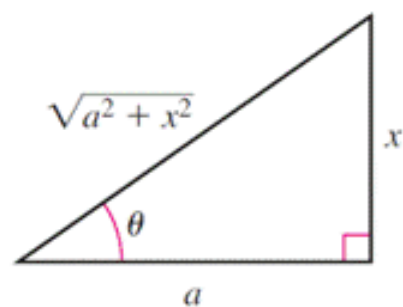
$$x^2 - a^2$$

$$x = a \sec \theta, \begin{cases} 0 \leq \theta < \frac{\pi}{2} & \text{for } x \geq a \\ \frac{\pi}{2} < \theta \leq \pi & \text{for } x \leq -a \end{cases}$$

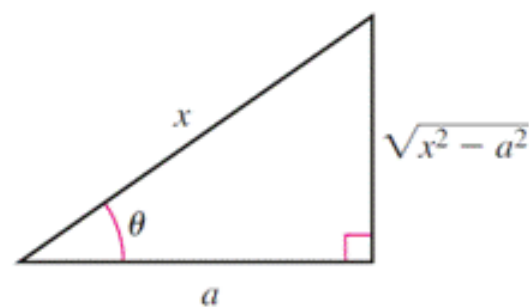
$$a^2 \sec^2 \theta - a^2 = a^2 \tan^2 \theta$$



$$x = a \sin \theta$$



$$x = a \tan \theta$$



$$x = a \sec \theta$$

FIGURE 8.5

Example 2

□ Find $\int \frac{1}{x^2 \sqrt{x^2 + 4}} dx$.

□ **Solution:**

Let $x = 2 \tan \theta$, $-\pi/2 < \theta < \pi/2$. Then $dx = 2 \sec^2 \theta d\theta$
and

$$\sqrt{x^2 + 4} = \sqrt{4(\tan^2 \theta + 1)} = \sqrt{4 \sec^2 \theta}$$

$$= 2 |\sec \theta|$$

$$= 2 \sec \theta$$

Example 2 – *Solution*

Thus we have

$$\begin{aligned}\int \frac{dx}{x^2 \sqrt{x^2 + 4}} &= \int \frac{2 \sec^2 \theta d\theta}{4 \tan^2 \theta \cdot 2 \sec \theta} \\ &= \frac{1}{4} \int \frac{\sec \theta}{\tan^2 \theta} d\theta\end{aligned}$$

To evaluate this trigonometric integral we put everything in terms of $\sin \theta$ and $\cos \theta$:

$$\frac{\sec \theta}{\tan^2 \theta} = \frac{1}{\cos \theta} \cdot \frac{\cos^2 \theta}{\sin^2 \theta}$$

Example 2 – *Solution*

$$= \frac{\cos \theta}{\sin^2 \theta}$$

Therefore, making the substitution $u = \sin \theta$, we have

$$\int \frac{dx}{x^2 \sqrt{x^2 + 4}} = \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta$$

$$= \frac{1}{4} \int \frac{du}{u^2}$$

$$= \frac{1}{4} \left(-\frac{1}{u} \right) + C$$

Example 2 – *Solution*

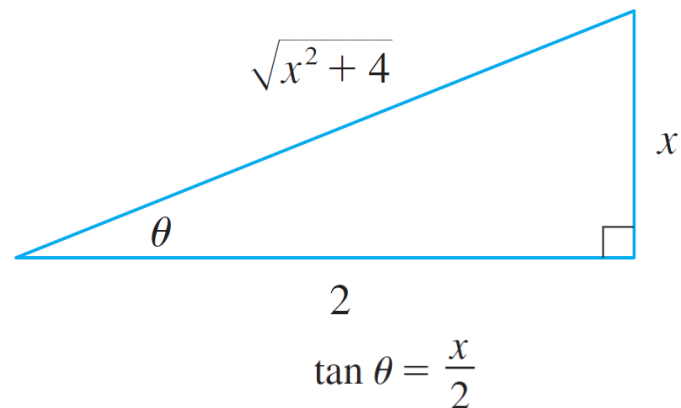
$$= -\frac{1}{4 \sin \theta} + C$$

$$= -\frac{\csc \theta}{4} + C$$

Example 2 – Solution

We use the figure below to determine that $\csc \theta = \sqrt{x^2 + 4}/x$ and so

$$\int \frac{dx}{x^2 \sqrt{x^2 + 4}} = -\frac{\sqrt{x^2 + 4}}{4x} + C$$



Example 3

Find $\int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx.$

Solution:

First we note that $(4x^2 + 9)^{3/2} = (\sqrt{4x^2 + 9})^3$, so trigonometric substitution is appropriate.

Although $\sqrt{4x^2 + 9}$ is not quite one of the expressions in the table of trigonometric substitutions, it becomes one of them if we make the preliminary substitution $u = 2x$.

Example 3 – Solution

When we combine this with the tangent substitution, we have $x = \frac{3}{2} \tan \theta$, which gives $dx = \frac{3}{2} \sec^2 \theta d\theta$ and

$$\begin{aligned}\sqrt{4x^2 + 9} &= \sqrt{9 \tan^2 \theta + 9} \\ &= 3 \sec \theta\end{aligned}$$

When $x = 0$, $\tan \theta = 0$, so $\theta = 0$; when $x = 3\sqrt{3}/2$, $\tan \theta = \sqrt{3}$, so $\theta = \pi/3$.

$$\int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx = \int_0^{\pi/3} \frac{\frac{27}{8} \tan^3 \theta}{27 \sec^3 \theta} \frac{3}{2} \sec^2 \theta d\theta$$

Example 3 – *Solution*

$$= \frac{3}{16} \int_0^{\pi/3} \frac{\tan^3 \theta}{\sec \theta} d\theta :$$

$$= \frac{3}{16} \int_0^{\pi/3} \frac{\sin^3 \theta}{\cos^2 \theta} d\theta$$

$$= \frac{3}{16} \int_0^{\pi/3} \frac{1 - \cos^2 \theta}{\cos^2 \theta} \sin \theta d\theta$$

- Now we substitute $u = \cos \theta$ so that $du = -\sin \theta d\theta$.
When $\theta = 0$, $u = 1$; when $\theta = \pi/3$, $u = \frac{1}{2}$.

Example 3 – *Solution*

Therefore

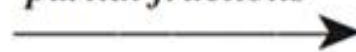
$$\begin{aligned}\int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx &= -\frac{3}{16} \int_1^{1/2} \frac{1 - u^2}{u^2} du \\&= \frac{3}{16} \int_1^{1/2} (1 - u^{-2}) du \\&= \frac{3}{16} \left[u + \frac{1}{u} \right]_1^{1/2} \\&= \frac{3}{16} \left[\left(\frac{1}{2} + 2 \right) - (1 + 1) \right] \\&= \frac{3}{32}\end{aligned}$$

Partial Fractions

rational function

$$\frac{3x}{(x-2)(x+4)} = \frac{3x}{x^2 + 2x - 8}$$

*method of
partial fractions*



*partial fraction
decomposition*

$$\frac{1}{x-2} + \frac{2}{x+4}$$

Difficult to integrate:

$$\int \frac{3x}{x^2 + 2x - 8} dx$$

Easy to integrate:

$$\int \left(\frac{1}{x-2} + \frac{2}{x+4} \right) dx$$

Integration of Rational Functions by Partial Fractions

To see how the method of partial fractions works in general, let's consider a rational function

$$f(x) = \frac{P(x)}{Q(x)}$$

where P and Q are polynomials.

It's possible to express f as a sum of simpler fractions provided that the degree of P is less than the degree of Q . Such a rational function is called *proper*.

Integration of Rational Functions by Partial Fractions

- If f is *improper*, that is, $\deg(P) \geq \deg(Q)$, then we must take the preliminary step of dividing Q into P (by long division) until a remainder $R(x)$ is obtained such that $\deg(R) < \deg(Q)$.

$$f(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$$

where S and R are also polynomials.

Example 1

Find $\int \frac{x^3 + x}{x - 1} dx.$

Solution:

Since the degree of the numerator is greater than the degree of the denominator, we first perform the long division.

This enables us to write:

$$\begin{aligned} \int \frac{x^3 + x}{x - 1} dx &= \int \left(x^2 + x + 2 + \frac{2}{x - 1} \right) dx \\ &= \frac{x^3}{3} + \frac{x^2}{2} + 2x + 2 \ln |x - 1| + C \end{aligned}$$

Integration of Rational Functions by Partial Fractions

If $f(x) = R(x)/Q(x)$ is a proper rational function:
factor the denominator $Q(x)$ as far as possible.

Ex: if $Q(x) = x^4 - 16$, we could factor it as

$$Q(x) = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$$

Integration of Rational Functions by Partial Fractions

Next: express the proper rational function as a sum of **partial fractions** of the form

$$\frac{A}{(ax + b)^i} \quad \text{or} \quad \frac{Ax + B}{(ax^2 + bx + c)^j}$$

A theorem in algebra guarantees that it is always possible to do this.

Four cases can occur.

Integration of Rational Functions by Partial Fractions

Case I The denominator $Q(x)$ is a product of distinct linear factors.

This means that we can write

$$Q(x) = (a_1x + b_1)(a_2x + b_2) \cdots (a_kx + b_k)$$

where no factor is repeated (and no factor is a constant multiple of another).

Integration of Rational Functions by Partial Fractions

In this case the partial fraction theorem states that there exist constants $A_1, A_2, \dots, A_k$ such that

$$\frac{R(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \dots + \frac{A_k}{a_kx + b_k}$$

These constants can be determined as in the next example.

Example 2

Evaluate $\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx.$

Solution:

Since the degree of the numerator is less than the degree of the denominator, we don't need to divide.

We factor the denominator as

$$2x^3 + 3x^2 - 2x = x(2x^2 + 3x - 2)$$

$$= x(2x - 1)(x + 2)$$

Example 2 – *Solution*

Since the denominator has three distinct linear factors, the partial fraction decomposition of the integrand has the form

$$\frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2}$$

To determine the values of A , B , and C , we multiply both sides of this equation by the product of the denominators, $x(2x - 1)(x + 2)$, obtaining

$$x^2 + 2x - 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1)$$

Example 2 – *Solution*

Expanding the right side and writing it in the standard form for polynomials, we get

$$x^2 + 2x - 1 = (2A + B + 2C)x^2 + (3A + 2B - C)x - 2A$$

These polynomials are identical, so their coefficients must be equal. The coefficient of x^2 on the right side, $2A + B + 2C$, must equal the coefficient of x^2 on the left side—namely, 1.

Likewise, the coefficients of x are equal and the constant terms are equal.

Example 2 – Solution

This gives the following system of equations for A , B , and C :

$$2A + B + 2C = 1$$

$$3A + 2B - C = 2$$

$$-2A = -1$$

□ Solving, we get, $A = \frac{1}{2}$, $B = \frac{1}{5}$, and $C = -\frac{1}{10}$, and so

$$\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx = \int \left[\frac{1}{2} \frac{1}{x} + \frac{1}{5} \frac{1}{2x - 1} - \frac{1}{10} \frac{1}{x + 2} \right] dx$$

Example 2 – *Solution*

$$= \frac{1}{2} \ln |x| + \frac{1}{10} \ln |2x - 1| - \frac{1}{10} \ln |x + 2| + K$$

In integrating the middle term we have made the mental substitution $u = 2x - 1$, which gives $du = 2 \, dx$ and $dx = \frac{1}{2} du$.

□ **Note:**

We can use an alternative method to find the coefficients A , B and C . We can choose values of x that simplify the equation:

$$x^2 + 2x - 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1)$$

If we put $x = 0$, then the second and third terms on the right side vanish and the equation then becomes $-2A = -1$, or $\frac{1}{2}A = \frac{1}{2}$.

Likewise, $x = \frac{1}{2}$ gives $5B/4 = \frac{1}{4}$ and $x = -2$ gives $10C = -1$, so $B = \frac{1}{5}$ and $C = -\frac{1}{10}$.

Integration of Rational Functions by Partial Fractions

Case II: $Q(x)$ is a product of linear factors, some of which are repeated.

Suppose the first linear factor $(a_1x + b_1)$ is repeated r times; that is, $(a_1x + b_1)^r$ occurs in the factorization of $Q(x)$. Then instead of the single term $A_1/(a_1x + b_1)$ in the equation:

$$\frac{R(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \cdots + \frac{A_k}{a_kx + b_k}$$

we use

$$\frac{A_1}{a_1x + b_1} + \frac{A_2}{(a_1x + b_1)^2} + \cdots + \frac{A_r}{(a_1x + b_1)^r}$$

Integration of Rational Functions by Partial Fractions

Example, we could write

$$\frac{x^3 - x + 1}{x^2(x - 1)^3} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x - 1} + \frac{D}{(x - 1)^2} + \frac{E}{(x - 1)^3}$$

Example 3

Find $\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx.$

Solution:

The first step is to divide. The result of long division is

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = x + 1 + \frac{4x}{x^3 - x^2 - x + 1}$$

Example 3 – *Solution*

The second step is to factor the denominator

$$Q(x) = x^3 - x^2 - x + 1.$$

Since $Q(1) = 0$, we know that $x - 1$ is a factor and we obtain

$$x^3 - x^2 - x + 1 = (x - 1)(x^2 - 1)$$

$$= (x - 1)(x - 1)(x + 1)$$

$$= (x - 1)^2(x + 1)$$

Example 3 – *Solution*

Since the linear factor $x - 1$ occurs twice, the partial fraction decomposition is

$$\frac{4x}{(x - 1)^2(x + 1)} = \frac{A}{x - 1} + \frac{B}{(x - 1)^2} + \frac{C}{x + 1}$$

Multiplying by the least common denominator, $(x - 1)^2(x + 1)$, we get

$$4x = A(x - 1)(x + 1) + B(x + 1) + C(x - 1)^2$$

Example 3 – *Solution*

$$= (A + C)x^2 + (B - 2C)x + (-A + B + C)$$

Now we equate coefficients:

$$A + C = 0$$

$$B - 2C = 4$$

$$-A + B + C = 0$$

Example 3 – *Solution*

Solving, we obtain $A = 1$, $B = 2$, and $C = -1$, so

$$\begin{aligned}\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx &= \int \left[x + 1 + \frac{1}{x - 1} + \frac{2}{(x - 1)^2} - \frac{1}{x + 1} \right] dx \\&= \frac{x^2}{2} + x + \ln |x - 1| - \frac{2}{x - 1} - \ln |x + 1| + K \\&= \frac{x^2}{2} + x - \frac{2}{x - 1} + \ln \left| \frac{x - 1}{x + 1} \right| + K\end{aligned}$$

Integration of Rational Functions by Partial Fractions

Case III: $Q(x)$ contains irreducible quadratic factors, none of which is repeated.

If $Q(x)$ has the factor $ax^2 + bx + c$, where $b^2 - 4ac < 0$, then, in addition to the partial fractions, the expression for $R(x)/Q(x)$ will have a term of the form

$$\frac{Ax + B}{ax^2 + bx + c}$$

where A and B are constants to be determined.

Integration of Rational Functions by Partial Fractions

- Example:

$f(x) = x/[(x - 2)(x^2 + 1)(x^2 + 4)]$ has the partial fraction decomposition:

$$\frac{x}{(x - 2)(x^2 + 1)(x^2 + 4)} = \frac{A}{x - 2} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{x^2 + 4}$$

- Any term of the form: $\frac{Ax + B}{ax^2 + bx + c}$ can be integrated by completing the square (if necessary) and using the formula

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$$

Example 4

Evaluate $\int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} dx.$

Solution:

Since the degree of the numerator is *not less than* the degree of the denominator, we first divide and obtain

$$\frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} = 1 + \frac{x - 1}{4x^2 - 4x + 3}$$

Example 4 – *Solution*

Notice that the quadratic $4x^2 - 4x + 3$ is irreducible because its discriminant is $b^2 - 4ac = -32 < 0$. This means it can't be factored, so we don't need to use the partial fraction technique.

To integrate the given function we complete the square in the denominator:

$$4x^2 - 4x + 3 = (2x - 1)^2 + 2$$

This suggests that we make the substitution $u = 2x - 1$.

Example 4 – *Solution*

Then $du = 2 \, dx$ and $x = \frac{1}{2}(u + 1)$, so

$$\int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} \, dx = \int \left(1 + \frac{x - 1}{4x^2 - 4x + 3} \right) \, dx$$

$$= x + \frac{1}{2} \int \frac{\frac{1}{2}(u + 1) - 1}{u^2 + 2} \, du$$

$$= x + \frac{1}{4} \int \frac{u - 1}{u^2 + 2} \, du$$

Example 4 – Solution

$$= x + \frac{1}{4} \int \frac{u}{u^2 + 2} du - \frac{1}{4} \int \frac{1}{u^2 + 2} du$$

$$= x + \frac{1}{8} \ln(u^2 + 2) - \frac{1}{4} \cdot \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{u}{\sqrt{2}}\right) + C$$

$$= x + \frac{1}{8} \ln(4x^2 - 4x + 3) - \frac{1}{4\sqrt{2}} \tan^{-1}\left(\frac{2x - 1}{\sqrt{2}}\right) + C$$

□ **Note:**

Example 6 illustrates the general procedure for integrating a partial fraction of the form

$$\frac{Ax + B}{ax^2 + bx + c} \quad \text{where } b^2 - 4ac < 0$$

We complete the square in the denominator and then make a substitution that brings the integral into the form

$$\int \frac{Cu + D}{u^2 + a^2} du = C \int \frac{u}{u^2 + a^2} du + D \int \frac{1}{u^2 + a^2} du$$

Then the first integral is a logarithm and the second is expressed in terms of $\tan^{-1}$.

Integration of Rational Functions by Partial Fractions

Case IV: $Q(x)$ contains a repeated irreducible quadratic factor.

If $Q(x)$ has the factor $(ax^2 + bx + c)^r$, where $b^2 - 4ac < 0$, then instead of the single partial fraction $\frac{Ax + B}{ax^2 + bx + c}$, the sum:

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}$$

occurs in the partial fraction decomposition of $R(x)/Q(x)$.

Each of the terms can be integrated by using a substitution or by first completing the square if necessary.

Example 5

Evaluate $\int \frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} dx.$

Solution:

The form of the partial fraction decomposition is

$$\frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{(x^2 + 1)^2}$$

Multiplying by $x(x^2 + 1)^2$, we have

$$-x^3 + 2x^2 - x + 1 = A(x^2 + 1)^2 + (Bx + C)x(x^2 + 1) + (Dx + E)x$$

Example 5 – *Solution*

$$= A(x^4 + 2x^2 + 1) + B(x^4 + x^2) + C(x^3 + x) + Dx^2 + Ex$$

$$= (A + B)x^4 + Cx^3 + (2A + B + D)x^2 + (C + E)x + A$$

If we equate coefficients, we get the system

$$A + B = 0 \quad C = -1 \quad 2A + B + D = 2 \quad C + E = -1 \quad A = 1$$

which has the solution $A = 1$, $B = -1$, $C = -1$, $D = 1$ and $E = 0$.

Example 5 – *Solution*

Thus

$$\begin{aligned}\int \frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} dx &= \int \left(\frac{1}{x} - \frac{x + 1}{x^2 + 1} + \frac{x}{(x^2 + 1)^2} \right) dx \\&= \int \frac{dx}{x} - \int \frac{x}{x^2 + 1} dx - \int \frac{dx}{x^2 + 1} + \int \frac{x dx}{(x^2 + 1)^2} \\&= \ln |x| - \frac{1}{2} \ln(x^2 + 1) - \tan^{-1} x - \frac{1}{2(x^2 + 1)} + K\end{aligned}$$

SUMMARY Partial Fraction Decompositions

Let $f(x) = p(x)/q(x)$ be a proper rational function in reduced form. Assume the denominator q has been factored completely over the real numbers and m is a positive integer.

- 1. Simple linear factor** A factor $x - r$ in the denominator requires the partial

fraction $\frac{A}{x - r}$.

- 2. Repeated linear factor** A factor $(x - r)^m$ with $m > 1$ in the denominator requires the partial fractions

$$\frac{A_1}{(x - r)} + \frac{A_2}{(x - r)^2} + \frac{A_3}{(x - r)^3} + \cdots + \frac{A_m}{(x - r)^m}.$$

- 3. Simple irreducible quadratic factor** An irreducible factor $ax^2 + bx + c$ in the denominator requires the partial fraction

$$\frac{Ax + B}{ax^2 + bx + c}.$$

- 4. Repeated irreducible quadratic factor** (See Exercises 67–70.) An irreducible factor $(ax^2 + bx + c)^m$ with $m > 1$ in the denominator requires the partial fractions

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_mx + B_m}{(ax^2 + bx + c)^m}.$$